

Potential of Bench Press Throw Performance Using a Heavy Load and Velocity-Based Repetition Control

Athanasios Tsoukos,¹ Lee E. Brown,² Gerasimos Terzis,¹ Panagiotis Veligekas,¹ and Gregory C. Bogdanis¹

¹School of Physical Education and Sports Science, National & Kapodistrian University of Athens, Athens, Greece; and ²Human Performance Laboratory, California State University, Fullerton, Fullerton, California

Abstract

Tsoukos, A, Brown, LE, Terzis, G, Veligekas, P, and Bogdanis, GC. Potential of bench press throw performance using a heavy load and velocity-based repetition control. *J Strength Cond Res* 35(2S): S72–S79, 2021—The acute effects of heavy load bench press exercise on subsequent bench press throw (BPT) performance and surface electromyographic (sEMG) activity were examined using movement velocity control. Eleven resistance-trained men completed 3 conditions in randomized and counter-balanced order. In 2 conditions, bench press was performed as fast as possible against a load of 80% of 1 repetition maximum, until mean velocity dropped to 90% (C90) or 70% (C70) of the fastest repetition. In the control condition, no bench press was performed. Single maximal BPT efforts were performed in all conditions after warm-up and throughout the 12 minutes of subsequent recovery. Mean propulsive velocity (MPV), peak velocity (PV), and sEMG activity of the prime mover muscles were measured during the BPT efforts. The total number of repetitions and volume load during bench press were significantly greater in C70 compared with C90 (6.5 ± 1.9 vs. 3.4 ± 0.9 repetitions and 557.8 ± 177.8 vs. 293.8 ± 95.7 kg, $p < 0.01$). In C90, MPV was increased 5.3–7.4% ($p < 0.01$) for a prolonged period (4–12 minutes), while in C70, MPV increased only at 10 minutes (+5.9%, $p < 0.01$) and 12 minutes (+4.3%, $p < 0.01$). Peak velocity was improved only in the C90 at 8–12th min by 3.6–4.7% ($p \leq 0.05$). Surface electromyographic activity of the pectoralis major muscle was significantly greater than baseline at 10 minutes of recovery only in the C90 ($p < 0.01$). Performance gains may be optimized by taking into account the individual fatigue profiles, allowing a percentage drop of only 10% in movement velocity during the conditioning bench press exercise.

Key Words: velocity loss, EMG activity, fatigue, PAP, PAPE, linear encoder

Introduction

Performance of previous resistance exercise may enhance subsequent muscle power output (14,28,36,38). This phenomenon has been termed “postactivation potentiation” (PAP) or “post-activation performance enhancement” (PAPE), and it has been researched extensively in recent years (3,36). Sports coaches are taking advantage of conditioning resistance exercises, to acutely enhance their athletes performance during competition or maximize training gains (4,19). Conditioning protocols commonly use a higher load resistance exercise to activate the neuromuscular system and induce an increase in performance of an explosive exercise with low load in the next 1–12 minutes (4,17,35). However, resistance exercise may also induce fatigue in parallel with neuromuscular activation (23), and if fatigue is substantial, performance will decrease rather than increase (5,12). The interaction between neuromuscular activation and fatigue is affected by the volume and the intensity of the conditioning exercise (8,28), the recovery period between activation and performance (18,37), the type of the conditioning exercise (5), and subjects’ characteristics, such as muscle fiber type and fatigability (12,35).

The load used for the conditioning exercise plays an important role in determining the magnitude of improvement of subsequent performance (20,38). Although some authors reported that

conditioning exercises with low and moderate loads (25–60% of 1 repetition maximum [1RM]) significantly improve subsequent muscle performance (31,32,38), there is evidence to suggest that higher loads (80–90% of 1RM) may have greater effects when compared with lower or moderate loads (20). However, the number of repetitions used with such loads may range from 3 to 8, depending on 2 variables: (a) the actual load used and (b) the fatigue profile of the subject. The number of repetitions and the concomitant level of fatigue when using heavy loads may influence subsequent performance, and thus, an individualized approach is necessary for performance optimization.

Surface electromyography has been used by researchers to examine the neural aspects of muscle activation and fatigue during conditioning activities (11,15,34), but the results are conflicting, with some studies showing lack of change in sEMG activity (7), while others showed an increase in sEMG activity that peaked at different time points after the conditioning exercise in different subjects (30).

Velocity-based resistance training has been used as a practical method for monitoring and quantifying the intensity and volume load of resistance exercise (21,22). Using this method, fatigue during a conditioning exercise may be monitored and controlled, so that to performance optimization is individualized (38). During velocity-based resistance training, a linear position encoder, a linear velocity transducer, or an accelerometer is used to monitor the peak and mean velocity of the barbell in every repetition (1). Recording of the velocity drop from the first to the

Address correspondence to Gregory C. Bogdanis, gbogdanis@phed.uoa.gr.

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last repetition may serve as a sensitive indicator of neuromuscular fatigue (10,25). Thus, velocity-based resistance training may be used to control the effects of fatigue during the conditioning exercise and, doing so, to optimize subsequent muscle performance. In a recent study, it was shown that bench press throw (BPT) performance is significantly improved by 4.5–6.8% after 4–12 minutes of bench press when mean velocity is not allowed to drop more than 10% of that attained in the first repetition (38). In that study, only light loads were used during the conditioning exercise (40 and 60% of 1RM), while there is no information regarding the effects of velocity controlled resistance exercise using heavy loads on neuromuscular activation. Previous research has shown that heavier loads may be more effective than lighter loads, when they are used as a conditioning exercise (20). Also, by equating fatigue between subjects using velocity-based repetition control, the same level of fatigue will be reached with a different number of repetitions, depending on the strength level and the fatigue characteristics of the muscles involved (17,35,38). Taking into consideration the above, the purpose of this study was to examine the acute effects of a heavy load bench press exercise on subsequent BPT and surface sEMG of the prime mover muscles, using movement velocity control. It was hypothesized that performance gains will be maximized by taking into account the individual fatigue profiles, allowing a percentage drop of 10% compared with 30% in movement velocity during the conditioning bench press exercise.

Methods

Experimental Approach to the Problem

A repeated-measures design was used to examine the acute effects of bench press exercise on subsequent BPT performance and sEMG activity. The study included 3 conditions: 2 experimental and 1 control (CTRL), in a randomized and counterbalanced order. In the 2 experimental conditions, the subjects were required to perform bench press repetitions as fast as possible against a load equal to 80% of 1RM, until mean movement velocity dropped by 10% (i.e., to 90% of the highest attained mean velocity) or by 30% (i.e., to 70% of the highest attained mean velocity) (Figure 1). The following dependent variables of the BPT exercise were assessed at 0.75, 2, 4, 6, 8, 10, and 12 minutes of recovery (time points) after the bench press exercise: (a) mean propulsive velocity (MPV), (b) peak velocity (PV), and (c, d) root mean square (RMS) of the pectoralis major and triceps brachii muscles normalized as a percentage of 1RM sEMG activity.

Subjects

Power analysis indicated a minimum sample size of 8 subjects would be needed to detect an effect size (ES) of 0.4. This value was obtained from the meta-analysis of Wilson et al. (40) regarding the effects of conditioning exercise on muscle power. Power analysis was performed using the following parameters: Type of analysis was set to repeated-measures analysis of variance (ANOVA), the required power was set to 0.80, alpha was set to 0.05, and the correlation between repeated measures was set to $r = 0.5$ (G-Power software, v.3.1.9.2).

Eleven men with a background in strength and power training of at least 3 years, who were involved in recreational (gym training) and team sports training (handball, soccer, and basketball), took part in the study (mean $\pm$ SD: age: 26.366.1 years [age range: 20–33 years], height: 1.83 ± 0.05 m, body mass: 84.8 ± 12.7 kg, % of

body fat: 13.7 ± 5.6 , 1RM bench press: 109.1 ± 23.0 kg, relative strength: 1.29 ± 0.23 kg·kg⁻¹). The study was conducted during the start of the transition period. During this period, subjects were training 3 days per week, were not taking any nutritional supplements or drugs, and were free from musculoskeletal injuries for at least 1 year before the study. Subjects were instructed to replicate their dietary intake for 24 hours before each testing session. After a thorough explanation of the testing protocol, the possible risks involved and the right to cease the participation at will, written informed consent was obtained from each subject. The study was approved by the institutional review board at the School of P.E. & Sport Science, National and Kapodistrian University of Athens (Approval no. 1084/13-10-2018), and all procedures were in accordance with the Code of Ethics of the World Medical Association (Helsinki declaration of 1964, as revised in 2013).

Procedures

Familiarization and Preliminary Measurements. The subjects took part in 3 preliminary sessions. In the first preliminary visit, subjects were familiarized with the BPT exercise and anthropometric data were obtained. Body height was measured to the nearest 0.1 cm using a stadiometer (Charder HM-200P Portstad). Body mass was measured to the nearest 0.1 kg by a scale (TBF-300A Body Composition Analyzer-Tanita), and body fat was calculated from 7 skinfold thicknesses (16) using a Harpenden skinfold calliper (British Indicators Ltd., Herts, England). The maximum dynamic bench press strength (1RM) was measured in the second preliminary visit, whereas in the third visit, subjects were familiarized in the conditioning bench press exercise and BPT. On that visit, they performed 2 separate sets of the bench press exercise, until mean velocity dropped to 90% or 70% of the peak attained (C90 or C70), against a resistance equal to 80% of 1RM. When mean velocity dropped to 90% or 70% of the peak attained, an audible beep was heard from a linear position transducer (Tendo Power Analyzer System v. 314; TENDO Sports Machines, Trencin, Slovak Republic) and the set was immediately terminated. The subjects were instructed to move the barbell as fast as possible during all phases of movement from the first repetition. After 1, 4, and 7 minutes from the end of each set, single repetitions of BPT were performed against a resistance of 30% 1RM, to familiarize subjects with the procedures of the experimental and control conditions.

General and Specific Warm-up. The subjects performed a standardized warm-up before each preliminary or experimental condition, which consisted of 5 minutes of light cycling on a cycle ergometer (50–60 watt) and 5 minutes of dynamic stretching of the arm and chest muscles. The 2 specific warm-ups were conducted after the general warm-up. The first was conducted before the baseline measurements of BPT and the other before performing the conditioning bench press exercise (Figure 1). The first specific warm-up (SWU 1) included 2 sets of BPT, against a resistance of 15% 1RM (6 repetitions) and 30% 1RM (4 repetitions) interspersed with 3 minutes of rest in the sitting position. The second specific warm-up (SWU 2) included 1 set of 8 repetitions using 40% 1RM and 1 set of 5 repetitions using 60% 1RM, with 3-minute rest in between in the sitting position.

Maximum Dynamic Strength Test (1 Repetition Maximum). Maximum dynamic strength (1RM) was assessed in the second preliminary visit on a Smith machine according to the procedures outlined by the National Strength and Conditioning Association (29).

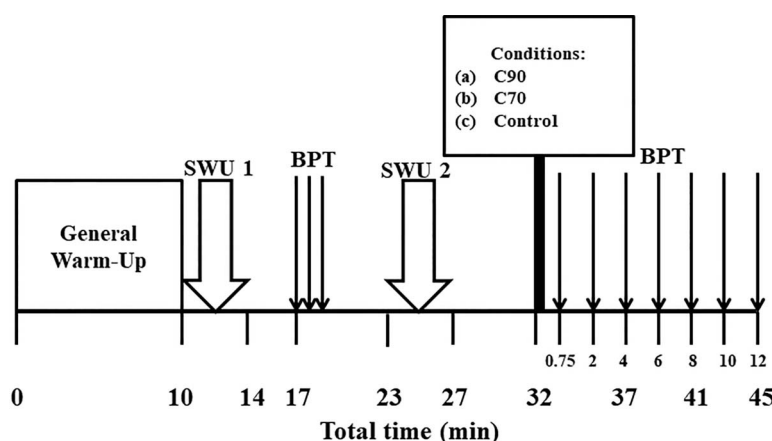


Figure 1. Schematic representation of the experimental protocol. BPT = bench press throw, SWU = specific warm-up.

The bench supported 3 body segments, the head, shoulders, and hips, and the feet were placed flat on the floor with a knee angle of approximately 90°. The subjects were instructed to use a closed pronated grip. Two experienced spotters who were strength and conditioning coaches were verbally encouraging the subjects and ensured their safety. The intraclass correlation coefficient (ICC) for the 1RM measurement was 0.92.

Experimental and Control Conditions. The schematic representation of the study design is shown in Figure 1. The 2 experimental conditions and the control condition (CTRL) were conducted 5–7 days apart in a random and counterbalanced order. The conditions were as follows:

- 80% of 1RM, subjects performed repetitions until mean velocity dropped to 90% of the highest attained (C90),
- 80% of 1RM, subjects performed repetitions until mean velocity dropped to 70% of the highest attained (C70), and
- Control condition in which subjects did not perform any conditioning activity (CA), but only the BPT (CTRL).

The subjects began each condition by performing the standardized general and specific warm-up 1 (Figure 1). After 3 minutes of recovery, they performed 3 BPTs on a Smith machine, with 30 seconds of recovery in between. This procedure was repeated in all conditions. The best value was kept as baseline. The specific warm-up 2 was performed 4 minutes later, and then, subjects rested for 5 minutes. Subsequently, they performed the conditioning bench press exercise or rested in the sitting position for 5 minutes (CTRL condition). After the conditioning bench press exercise, BPT performance was re-evaluated at the following time points: 45 seconds, 2, 4, 6, 8, 10, and 12 minutes.

Bench Press Throw. The BPT exercise was conducted on a Smith machine, using a load of 30% of 1RM. Subjects were instructed to hold the bar using the same grip as in bench press exercise, with the elbows fully extended. From this starting position, they performed a fast countermovement and threw the bar as fast and as high as possible (39). Subjects were assisted to stop the movement at the desired elbow angle of at least 90° by a “beep” sound that was activated when a photocell beam was interrupted by the barbell when elbow angle was just below 90°. The subjects kept their heads and core in contact with the bench and their feet on the floor and were asked to accelerate the barbell until the end of the

range of motion (39). After the execution of the BPT, 2 spotters caught and lowered the loaded bar to ensure the safety of the subjects.

Measurement of Movement Velocity. A linear position transducer (Tendo Power Analyzer System v. 314; TENDO Sports Machines) measured the movement velocities. During the bench press exercise, the mean and the peak velocity was monitored in every repetition. During the BPT, the MPV and PV during the concentric phase were calculated and measured, respectively. Mean propulsive velocity has been defined as the mean velocity from the beginning of the concentric phase of the repetition until the instant at which PV is reached (26). The ICCs for these measurements were as follows: MPV in BPT (0.945, $p < 0.001$, 95% confidence interval [CI]: 0.840–0.985, SEM: 0.02, %SEM: 1.6%, meaningful difference [MD]: 0.07, %MD: 4.4%) and PV in BPT (0.940, $p < 0.001$, 95% CI: 0.825–0.984, SEM: 0.03, %SEM: 1.5%, MD: 0.09, %MD: 4.2%).

Measurement of Surface Electromyographic Activity. Surface electromyographic activity of the pectoralis major (pars sternocostalis) and triceps brachii (lateral head) muscles of the right side of the body was recorded (Biopac MP35, Systems Inc., Santa Barbara, CA) and analyzed (Acqknowledge 4.2.0; Biopac Systems Inc.) as previously described (38). Bipolar Ag/AgCl electrodes were attached to the skin (inter-electrode distance: 20 mm) according to the recommendations of SENIAM (13). The ground electrode was placed on the clavicle. The surface of the skin was shaved, cleaned with alcohol, and rubbed with fine sandpaper to keep the impedance low between the 2 electrodes. The sampling frequency of sEMG recordings was 1,000 Hz, and the signal was amplified (gain = 1,000) and filtered (band pass = 10–500 Hz). Surface electromyographic activity of the muscles was calculated as the RMS value between the onset and the end of the burst. The onset of the burst was defined when the mean baseline sEMG activity exceeded the threshold value of 3 SDs. The RMS values of both muscles during BPTs were normalized to the RMS values during 1RM bench press (27,38). The ICC for the sEMG activity measurement for pectoralis major and triceps brachii muscles was 0.984 ($p < 0.001$, 95% CI: 0.953–0.996, SEM: 0.02, %SEM: 7.1%, MD: 0.05, %MD: 19.8%) and 0.993 (95% CI: 0.980–0.998, SEM: 0.03, %SEM: 7.0%, MD: 7.0%, %MD: 19.4%), respectively.

Statistical Analyses

Statistical analyses were conducted by using the SPSS Statistics Ver. 23 (IBM Corporation). All data are presented as means and SDs. The acute effects of the conditioning bench press exercise on the dependent variables were examined by two-way repeated-measures ANOVA (3 conditions $\times$ 8 time points). Also, the highest MPV and PV for each subject, irrespective of recovery time, was compared with the respective baseline (3 conditions $\times$ 2 time points). When a significant main effect or interaction was observed, a Tukey's post hoc test was performed. The ESs for main effects and interactions were determined by Partial eta-squared (η^2) values. Partial eta-squared (η^2) values were classified as small (0.01–0.059), moderate (0.06–0.137), and large (>0.137). For pairwise comparisons, the ES was determined by Hedges' g (small, <0.3 ; medium, 0.3 – 0.8 ; large, >0.8). The ICC was calculated using a two-way mixed model to determine test-retest reliability for all dependent variables. Relationships between variables were obtained by calculating Pearson product-moment correlation coefficient. Statistical significance was set at $p \leq 0.05$.

Results

Mean and Peak Movement Velocity During Bench Press

In the experimental conditions, the total number of repetitions during the bench press exercise was significantly different between conditions (Table 1) ($t = 7.1$; $p < 0.001$; Hedges' $g = 2.0$). Volume load, calculated as the product of the load lifted multiplied by the number of repetitions, was also significantly different between conditions ($t = 7.5$; $p < 0.001$; Hedges' $g = 1.8$).

Mean velocity of the first and last repetitions in the bench press exercise during the 2 experimental conditions is presented in Figure 2A. The 2-way ANOVA (3 conditions $\times$ 2 time points) showed a significant interaction ($p < 0.01$; $\eta^2 = 0.82$). Tukey's post hoc tests indicated a significant drop in mean velocity from the first to the last repetition for both C90 ($p < 0.01$; Hedges' $g = 1.0$) and C70 ($p < 0.01$; Hedges' $g = 2.2$) (Figure 2). Also, there was a significant difference in mean velocity between the last repetitions for C90 and C70 ($p < 0.01$; Hedges' $g = 1.4$). No difference was observed between the first repetitions for C90 and C70 in the mean velocity ($p > 0.05$).

Peak velocity of the first and last repetitions in the bench press exercise during the CA is presented in Figure 2B. The 2-way ANOVA (3 conditions $\times$ 2 time points) showed a significant interaction ($p < 0.01$; $\eta^2 = 0.63$). Tukey's post hoc tests indicated a significant drop in PV from the first to the last repetition for both C90 ($p < 0.01$; Hedges' $g = 0.9$) and C70 ($p < 0.01$; Hedges' $g = 2.5$) (Figure 2). There was a significant difference in PV between the last repetitions for C90 and C70 ($p < 0.01$; Hedges' $g = 1.4$). No difference was observed between the first repetitions for C90 and C70 in the PV ($p > 0.05$).

Time Course of Mean Propulsive Velocity During the Bench Press Throw Exercise

The time course of changes in MPV performance during the experimental and control conditions is shown in Figure 3. Baseline values were similar in all conditions ($p = 0.99$). The 2-way ANOVA (3 conditions $\times$ 8 time points) revealed a significant condition $\times$ time interaction ($p < 0.001$, $\eta^2 = 0.41$). Mean

Table 1

Characteristics of the conditioning activity (bench press using 80% of 1 repetition maximum) in the trials when mean velocity was allowed to drop to 90% (C90) and 70% (C70) of the maximum attained.*

Variables	Conditions	
	C90	C70
Lifted load (kg)	87.3 $\pm$ 18.4	87.3 $\pm$ 18.4
No. of repetitions	3.4 $\pm$ 0.9	6.5 $\pm$ 1.9†
Volume load (kg)	293.8 $\pm$ 95.7	557.8 $\pm$ 177.8†
Max no. of repetitions	5	10
Min no. of repetitions	2	4

*Values are presented as mean $\pm$ SD. Max and min number of repetitions refer to the highest and lowest values recorded in the individual data.

† $p < 0.01$ compared with C90.

propulsive velocity performance was improved in the C90 condition from the fourth ($+5.3 \pm 2.8\%$, $p < 0.01$, Hedges' $g = 0.60$) to the 12th minute of recovery ($+6.1 \pm 2.7\%$, $p < 0.01$, Hedges' $g = 0.70$). The highest MPV was observed on the 10th minute of recovery ($+7.4 \pm 2.1\%$, $p < 0.01$, Hedges' $g = 0.9$). By contrast, MPV in the C70 condition was improved only at the 10th minute ($+5.9 \pm 4.9\%$, $p < 0.01$, Hedges' $g = 0.7$) and the 12th minute of recovery ($+4.3 \pm 4.3\%$, $p < 0.01$, Hedges' $g = 0.6$). Also, a significant drop in MPV performance from baseline

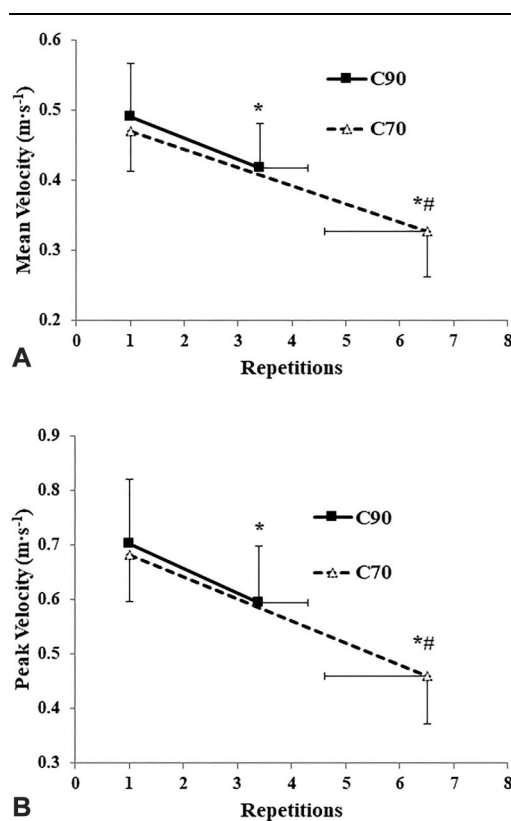


Figure 2. Mean (A) and peak (B) velocity of the first and last repetitions during the conditioning bench press exercise across the experimental and control conditions, plotted against the number of repetitions. Values are presented as mean $\pm$ SD. * $p < 0.01$ between the first and the last repetition in the C90 and C70 conditions, respectively. # $p < 0.01$ between the last repetitions of C90 and C70 conditions, respectively.

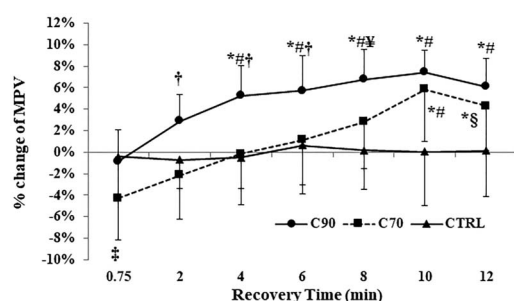


Figure 3. Time course of changes of mean propulsive velocity (MPV) during the bench press throws. Values are expressed as percent changes compared with baseline values. * $p < 0.01$ compared with baseline, † $p < 0.05$ compared with baseline, # $p < 0.01$ compared with the CTRL condition, ‡ $p < 0.01$ compared with the C70 condition, ¥ $p < 0.05$ compared with the C70 condition, § $p < 0.05$ compared with the CTRL condition.

was observed immediately after the conditioning bench press exercise in the C70 condition by $-4.3 \pm 3.9\%$ ($p < 0.05$; Hedges' $g = 0.6$).

Furthermore, MPV during the C90 condition was greater compared with (a) the corresponding values during the CTRL condition from the fourth ($p < 0.001$, Hedges' $g = 2.0$) to the 12th minute of recovery ($p < 0.001$, Hedges' $g = 1.6$) and (b) the corresponding value during the C70 condition from the second ($p < 0.001$, Hedges' $g = 1.4$) to the eighth minute of recovery ($p < 0.05$, Hedges' $g = 1.0$).

Mean propulsive velocity was significantly greater in the C70 condition compared with the corresponding value in the CTRL condition in the 10th ($p < 0.01$, Hedges' $g = 0.7$) and in the 12th minute of recovery ($p < 0.05$, Hedges' $g = 0.6$).

The best MPV for each subject irrespective of recovery time in the C90 condition was $8.5 \pm 2.4\%$ greater than baseline ($p < 0.01$, Hedges' $g = 1.0$) and was observed on the 9.8 ± 1.9 minutes of recovery. Also, the best MPV during recovery in the C90 condition was greater compared with the CTRL ($p < 0.01$, Hedges' $g = 0.6$).

The best MPV for each subject irrespective of recovery time in the C70 condition was $7.0 \pm 4.5\%$ greater than baseline ($p < 0.01$, Hedges' $g = 0.9$) and was observed at the 9.6 ± 1.5 minutes of recovery. Also, the best MPV during recovery in the C70 condition was greater compared with the CTRL ($p < 0.01$, Hedges' $g = 0.4$). No difference was found between the C90 and C70 ($p > 0.05$) for the best individual MPV.

Time Course of Peak Velocity During the Bench Press Throw Exercise

The time course of changes in PV during the experimental and control conditions is shown in Figure 4. Baseline values were similar between the conditions ($p > 0.05$). The 2-way ANOVA (3 conditions $\times$ 8 time points) revealed a significant condition $\times$ time interaction ($p < 0.001$, $\eta^2 = 0.32$). Tukey's post hoc tests indicated that PV was increased compared with baseline only in the C90 condition from the eighth ($+4.2 \pm 3.9\%$, $p < 0.01$, Hedges' $g = 0.6$) to the 12th minute of recovery ($+3.6 \pm 1.8\%$, $p = 0.05$, Hedges' $g = 0.6$). The highest PV was observed on the 10th minute of recovery ($+4.7 \pm 2.6\%$, $p < 0.01$, Hedges' $g = 0.7$). Furthermore, PV during the C90 condition was greater compared with (a) the corresponding value during the CTRL

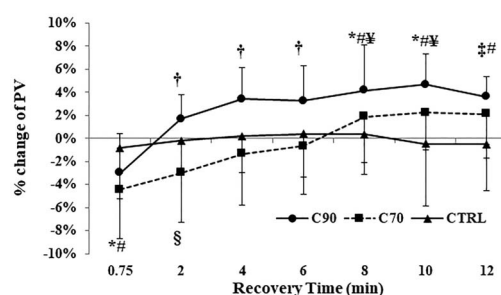


Figure 4. Time course of changes of peak velocity (PV) during the bench press throws. Values are expressed as percent changes compared with baseline values. * $p < 0.01$ compared with baseline, † $p < 0.05$ compared with baseline, # $p < 0.01$ compared with the CTRL condition, ‡ $p < 0.01$ compared with the C70 condition, ¥ $p < 0.05$ compared with the C70 condition, § $p < 0.05$ compared with the CTRL condition.

condition from the eighth ($p = 0.01$, Hedges' $g = 0.4$) to the 12th minute of recovery ($p < 0.001$, Hedges' $g = 0.5$) and (b) the corresponding value during the C70 condition from the second ($p < 0.001$, Hedges' $g = 0.8$) to the 10th minute of recovery ($p < 0.05$, Hedges' $g = 0.5$).

In contrast, PV was decreased in the C70 condition immediately after the conditioning bench press exercise compared with baseline (0.75 minutes) by $-4.4 \pm 4.3\%$ ($p < 0.01$; Hedges' $g = 0.6$). Also PV was significantly lower compared with the corresponding time point during the CTRL condition at the 0.75 ($p < 0.01$; Hedges' $g = 0.5$) and 2 minutes of recovery ($p < 0.05$; Hedges' $g = 0.4$).

The best PV for each subject irrespective of recovery time in the C90 condition was $6.0 \pm 2.1\%$ greater than baseline ($p < 0.01$, Hedges' $g = 0.90$) and was observed at the 8.5 ± 3.1 minutes of recovery. Also, the best PV during recovery of the C90 condition was greater compared with (a) the CTRL condition ($p < 0.01$, Hedges' $g = 0.5$) and (b) the C70 condition ($p < 0.01$, Hedges' $g = 0.4$).

The best PV for each subject irrespective of recovery time in the C70 condition was $4.0 \pm 3.7\%$ greater than baseline ($p < 0.01$, Hedges' $g = 0.6$) and was observed at the 9.3 ± 2.1 minutes of recovery. Also, the best PV during recovery of the C70 condition was not greater compared with the CTRL ($p > 0.01$).

Time Course of Surface Electromyographic Activity of the Pectoralis Major and Triceps Brachii Muscles

The 2-way ANOVA (3 conditions $\times$ 8 time points) showed a significant 2-way interaction (condition $\times$ time point, $p = 0.03$, $\eta^2 = 0.22$). Tukey's post hoc tests revealed that the sEMG activity of pectoralis major muscle during the C90 condition was greater when compared with baseline, on the 10th minute of recovery ($p < 0.001$, Hedges' $g = 0.7$). This value was greater compared with the corresponding value during the C70 condition ($p < 0.001$, Hedges' $g = 0.7$) and the corresponding value during the CTRL condition ($p = 0.03$, Hedges' $g = 0.5$) (Table 2).

A significant 2-way interaction was also observed in the sEMG activity (RMS) of pectoralis major muscle activity during the best MPV, irrespective of time point of recovery (3 conditions $\times$ 8 time points, $p = 0.03$, $\eta^2 = 0.44$). Tukey's post hoc tests indicated that the sEMG activity (RMS) of pectoralis major muscle was greater in the C90 during the best MPV, compared with the baseline ($p < 0.01$, Hedges' $g = 0.7$), with the C70 ($p < 0.05$, Hedges' $g = 0.5$)

Table 2**Time course of changes of sEMG activity of pectoralis major muscle.*†**

Time points (min)	EMG activity of the pectoralis major		
	Condition		
	C90	C70	CTRL
PRE	30 ± 11	30 ± 12	32 ± 12
0.75	32 ± 13	26 ± 9	32 ± 12
2	34 ± 16	26 ± 11	31 ± 12
4	36 ± 15	26 ± 11	30 ± 10
6	34 ± 13	31 ± 13	36 ± 13
8	39 ± 18	33 ± 16	32 ± 13
10	43 ± 22‡§	31 ± 11	33 ± 13
12	39 ± 20	32 ± 12	35 ± 13
BEST	42 ± 22‡ ¶	33 ± 15	34 ± 13

*sEMG = surface electromyography.

†Values are normalized as a percentage to 1RM sEMG activity and expressed as mean ± SD. The BEST time point refers to the sEMG activity of the highest mean propulsive velocity during recovery.

‡ $p < 0.01$ compared with baseline.§ $p < 0.01$ compared with the corresponding value in the C70 condition.|| $p < 0.05$ compared with the corresponding value in the CTRL condition.¶ $p < 0.05$ compared with the corresponding value in the C70 condition.

and with the CTRL conditions ($p < 0.05$, Hedges' $g = 0.4$). No difference was found between the C70 and CTRL conditions ($p > 0.05$).

The 2-way ANOVA (3 conditions $\times$ 8 time points) showed a nonsignificant 2-way interaction for the triceps brachii muscle sEMG during the time course of recovery (condition $\times$ time point, $p = 0.43$, $\eta^2 = 0.17$) as well as when peak individual responses were taken into account (3 conditions $\times$ 2 time points, $p = 0.59$, $\eta^2 = 0.10$) (Table 3).

Discussion

The main finding of this study was that MPV during BPT acutely improved by a similar magnitude (7–8%) in both conditions, i.e., when mean velocity during the conditioning bench press exercise dropped to 90 and 70% of the maximum (Figures 3 and 4). However, in the C90 condition, MPV during BPT was increased from the early recovery (2 minutes) and remained high for 10 minutes (Figure 3). By contrast, when fatigue during the conditioning bench press exercise was allowed to develop to a greater extent (C70), the increase in MPV manifested only after 10 minutes of recovery. A similar pattern was observed for PV, which was increased by 5–6% from the eighth minute onward, only in the C90 condition, but remained unchanged in the C70 condition. **Thus, it may be argued that fatigue may counteract the potentiating effect of the conditioning bench press exercise, and this can be individually controlled by allowing only a small and individualized drop in movement velocity (i.e., 10% of the individual maximum speed).**

The interaction between potentiation and fatigue in determining neuromuscular performance after a conditioning exercise is well known (23). Since the load was the same in the 2 experimental conditions (i.e., 80% 1RM), the number of repetitions performed determined the volume load and the development of fatigue. **Thus, in the C90 condition, where volume load (load $\times$ repetitions) was lower, fatigue was smaller and performance potentiation was evident for a wide, 8 minutes, time window (fourth to the 12th minute of recovery).** The role of volume load on performance potentiation has been shown in

Table 3**Time course of changes of sEMG activity of triceps brachii muscle.*†**

Time points (min)	EMG activity of the triceps brachii		
	Condition		
	C90	C70	CTRL
PRE	33 ± 22	37 ± 27	38 ± 32
0.75	33 ± 25	31 ± 36	35 ± 25
2	37 ± 30	36 ± 24	35 ± 23
4	35 ± 22	42 ± 27	39 ± 26
6	36 ± 25	40 ± 23	38 ± 28
8	39 ± 32	38 ± 17	36 ± 27
10	38 ± 28	41 ± 22	39 ± 28
12	39 ± 28	43 ± 23	35 ± 24
BEST	40 ± 28	39 ± 15	36 ± 23

*sEMG = surface electromyography.

†Values are normalized as a percentage to 1RM sEMG activity and expressed as mean ± SD. The BEST time point refers to the sEMG activity of the highest mean propulsive velocity during recovery.

previous studies, that reported significant improvements of performance by using lower than higher volume load during CAs (28,38). For example, in a similar study, Tsoukos et al. (38) examined the acute effects of the bench press exercise with low (40% of 1RM) and moderate loads (60% of 1RM), using 2 predetermined movement velocity loss percentages (mean velocity drop to 90% or 70% of the fastest repetition) on BPT performance and sEMG activity. The authors found that MPV and PV were increased from the fourth to the 12th minute of recovery by 4.5–6.8% only after the 60% 1RM condition during which velocity dropped to 90% and volume load was the lowest of all conditions (433 ± 131 kg). In this study, the volume load during the C90 was 293.8 ± 95.7 , whereas during the C70, it was about 90% higher (Table 1). Taken collectively, these results show the important role of total volume load and, indirectly, highlight the importance of fatigue to manifest improvements in explosive performance after a conditioning exercise. The development of fatigue during conditioning muscle contractions is well known from previous studies (12). In some studies aiming to potentiate muscle performance, the development of fatigue during the conditioning contractions is evident as a decrease in explosive performance below baseline levels during the first 4–6 minutes of the subsequent recovery (19,38). This decrease occurs when the number of repetitions is high and fatigue is substantial (38) and, in extreme cases, may even impair explosive performance for 6–24 hours later (22). However, in this study, there was a decrease in both MPV and PV by about 4% during the initial 4 minutes of recovery, despite the fact that the number of repetitions was low (i.e., <7 ; Table 1). Thus, to optimize the effects of a conditioning exercise on subsequent muscle power generation, one has to take into account not simply the number of repetitions performed against a certain load, but the level of fatigue experienced by each individual. In that respect, monitoring the changes in movement velocity during the conditioning exercise seems to be a promising method to achieve this outcome. Also, combining the results of this study with those from previous studies (38) leads to the conclusion that performance enhancement may be attained within a wide range of loads (from 40 to 80% 1RM), provided that the movement velocity during the conditioning exercise is allowed to drop by only by a small percentage of maximum (i.e., by 10%).

Another main finding was that sEMG activity of the pectoralis major muscle was greater on the 10th minute of

recovery as well as when the best individual responses were taken into account, only in the C90 condition (Table 2). This result is in agreement with previous studies reporting increased sEMG activity after conditioning muscle contractions (34,38) and suggests that during the C90 condition, part of the improvement in BPT performance may be related to neural mechanisms (e.g., increased recruitment of motor units or increased excitability of motoneurons) (28,36). However, other studies found no change in sEMG activity after maximal isometric conditioning contractions that enhanced BPT performance during recovery (7), while others reported a depression of sEMG activity for 30 minutes after performing a large volume of resistance exercise (30). This inconsistency in the changes in sEMG and performance is also evident in the C70 condition in this study, where there was no significant change in sEMG activity while BPT was increased. This implies that other mechanisms, in addition to neural factors, may also be important for performance enhancement, such as increased regulatory myosin light-chain phosphorylation, elevated muscle temperature, increased muscle-tendon stiffness, and intramuscular fluid accumulation (3,33,36).

During the C90 condition, the improvements in performance observed between the fourth and the 12th minute of recovery were greater than those reported in the literature. For example, Crewther et al. (6) found a 3–3.8% improvement in the vertical jump after a set of 3 RM back squats between the 4th and 12th minute of recovery and Bevan et al. (2) reported a 4.2% enhancement of BPT performance after 3 sets of 3 repetitions at 87% of 1RM (2) at the eighth minute of recovery. Also, West et al. (39) reported a 3.6% increase in peak power output of the arms 8 minutes after 3 sets of 3 repetitions of heavy (87% 1RM) conditioning bench press exercise. These improvements are less than half of what was found in this study in the C90 condition ($8.5 \pm 2.4\%$ in best individual MPV). This difference may be partly attributed to the use of velocity-based control of fatigue during the conditioning bench press exercise (10,25). When movement velocity was allowed to drop only by 10%, the fatigue levels were low, and performance enhancement of high magnitude was manifested for a considerable part of the recovery time (4–12 minutes). By contrast, when movement velocity was allowed to drop by 30%, fatigue was higher and an increase in performance was achieved later in the recovery (10–12 minutes) (9). PV was not improved over the control values throughout recovery in the C70 condition. However, there was an increase in MPV, suggesting that although PV was not improved, there was a higher rate of velocity development in the C70 condition. This is in agreement with the suggestion that maximum force or speed (i.e., the extremes of force-velocity relationship) is less influenced by conditioning contractions, while the rate of force and velocity development against submaximal loads are enhanced (24,36).

In conclusion, this study showed that a conditioning bench press set using a heavy load may bring about a substantial and long-lasting increase in subsequent BPT performance, provided that fatigue is controlled by measuring movement velocity loss. Furthermore, it was shown that power performance gains may be maximized by taking into account the individual fatigue profiles, allowing a percentage drop of only 10% in movement velocity during the conditioning bench press exercise. An increase of the volume load by allowing movement velocity to drop by 30% of the individual maximum

(C70 condition) results in a delay of performance enhancement and negates improvements in PV.

Practical Applications

The results of this study may help strength and conditioning coaches to prescribe effective complex training programs to acutely improve performance. This could be performed by monitoring the drop in movement velocity during complex training and thus individualizing the volume load of the conditioning exercise according to the fatigue profile. Allowing movement velocity of the conditioning bench press exercise to drop by only 10% of maximum will confer the greatest acute performance gains. Also, this study showed that in this type of velocity controlled training, coaches should record and use the individual time when peak performance is attained during recovery.

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